

Math 240 Fall 2023 - Homework 5 Solution

- (3.1.) 9. Compute the determinant by cofactor expansion. At each step, choose a row or column that involves the least amount of computation.

$$\begin{vmatrix} 4 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 3 & 0 & 0 & 0 \\ 8 & 3 & 1 & 7 \end{vmatrix}$$

Sol.

$$\begin{aligned} \begin{vmatrix} 4 & 0 & 0 & 5 \\ 1 & 7 & 2 & -5 \\ 3 & 0 & 0 & 0 \\ 8 & 3 & 1 & 7 \end{vmatrix} &= 3(-1)^{3+1} \begin{vmatrix} 0 & 0 & 5 \\ 7 & 2 & -5 \\ 3 & 1 & 7 \end{vmatrix} \\ &= 3 \left(5(-1)^{1+3} \begin{vmatrix} 7 & 2 \\ 3 & 1 \end{vmatrix} \right) \\ &= 15 \end{aligned}$$

- (3.2.) 6. Find the determinants by row reduction to echelon form.

$$\begin{vmatrix} 2 & 2 & -2 \\ 3 & 4 & -4 \\ 2 & -3 & -5 \end{vmatrix}$$

Sol.

$$\begin{aligned} \begin{vmatrix} 2 & 2 & -2 \\ 3 & 4 & -4 \\ 2 & -3 & -5 \end{vmatrix} &= 2 \begin{vmatrix} 1 & 1 & -1 \\ 3 & 4 & -4 \\ 2 & -3 & -5 \end{vmatrix} && \mathbf{r}_1 \rightarrow \frac{1}{2}\mathbf{r}_1 \\ &= 2 \begin{vmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & -5 & -3 \end{vmatrix} && \mathbf{r}_2 \rightarrow \mathbf{r}_2 - 3\mathbf{r}_1, \mathbf{r}_3 \rightarrow \mathbf{r}_3 - 2\mathbf{r}_1 \\ &= 2 \begin{vmatrix} 1 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -8 \end{vmatrix} && \mathbf{r}_3 \rightarrow \mathbf{r}_3 + 5\mathbf{r}_2 \\ &= 2(1)(1)(-8) = -16. \end{aligned}$$

The last line used the fact that determinant of a triangular matrix is given by the product of the diagonal entries.

- (3.3.) 24. Find the volume of the parallelepiped with one vertex at the origin and adjacent vertices at $(1, 3, 0)$, $(-2, 0, 2)$ and $(-1, 3, -1)$.

Sol. The volume of the parallelepiped is given by

$$\text{abs} \begin{pmatrix} \begin{vmatrix} 1 & -2 & -1 \\ 3 & 0 & 3 \\ 0 & 2 & -1 \end{vmatrix} \end{pmatrix} = \text{abs} \left(-3 \begin{vmatrix} -2 & -1 \\ 2 & -1 \end{vmatrix} - 3 \begin{vmatrix} 1 & -2 \\ 0 & 2 \end{vmatrix} \right) = |-18| = 18.$$